

Symbology and Quadrangular Polylogarithms for dS Cosmology

Yao-Qi Zhang

University of Hertfordshire

Based on works: [Generalised cluster adjacency](#) [2603.09965]

w. M. Capuano, L. Ferro, T. Łukowski, and A. Palazio

[Wavefunctions from quadrangular polylogarithms](#)

w. L. Ferro, T. Łukowski, S. Paranjape (loops), L. Ren, M. Spradlin, A. Volovich, and H.-C. Weng

Chain graphs: 2605.06542 ; loop graphs: to appear



chain graphs



loop graphs

Take-home message

Rudenko (in **type** *A*), and we (in **type** *B*), have defined:

Quadrangular Polylogarithms(QLi)

with manifest **total compatibility**
and applications to **cosmology**.

[Rudenko '20; Matveiakin, Rudenko '22] [w. L. Ferro, T. Łukowski, S. Paranjape et al., to appear.]

Plan of the talk

- Analytic structure & Total compatibility_[see Mattia and Allie's talk]
- Chain graphs & A -type QLi
- Loop graphs & B -type QLi

Polylogarithms and Symbology

MPL function

$$\text{Li}_{m_1, \dots, m_d}(z_1, \dots, z_d) = \sum_{0 < k_1 < \dots < k_d} \frac{z_1^{k_1} \dots z_d^{k_d}}{k_1^{m_1} \dots k_d^{m_d}}, \quad \text{weight } w = \sum_i m_i.$$

Iterated $d \log$ structure $\longrightarrow$ symbol words of length w [Goncharov, Spradlin, Vergu, Volovich '10]

$$\mathcal{S}(F^{(w)}) = \sum_{\alpha_1, \dots, \alpha_w} c_{\alpha_1 \dots \alpha_w} a_{\alpha_1} \otimes \dots \otimes a_{\alpha_w}, \quad \text{Symbol Alphabet } \mathcal{A} = \{a_\alpha\}.$$

$$d \text{Li}_2(z) = -\log(1-z) d \log z \implies \mathcal{S}(\text{Li}_2(z)) = -(1-z) \otimes z.$$

Planar $\mathcal{N} = 4$ SYM: six- and seven-point amplitude bootstrap

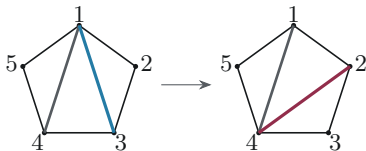
First-entry conditions + integrability + **cluster adjacency** + physical limits.

[Dixon, Drummond, Henn '11; Dixon et al. '16; Caron-Huot et al. '20; Dixon, Liu '23; He et al. '25]

Cluster algebras and cluster adjacency

Cluster algebra: a quiver with variables and mutation rules [See also Dani's Talk].

A_2 : triangulations are clusters



SYM: constrain consecutive letters

$$\cdots \otimes \underbrace{a_i \otimes a_{i+1}}_{\text{compatible pair}} \otimes \cdots$$

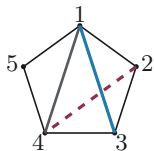
[Drummond, Foster, Gürdoğan '17]

A symbol word need *not* fit in one cluster.

$$\Delta_{13}\Delta_{24} = \Delta_{12}\Delta_{34} + \Delta_{14}\Delta_{23}.$$

Compatible variables share a cluster:
non-crossing diagonals for type A.

[Fomin, Zelevinsky '02]



$\Delta_{13} \otimes \Delta_{14} \otimes \Delta_{24}$ is allowed
the first and third chords cross

What happens when we bring these methods to cosmology?

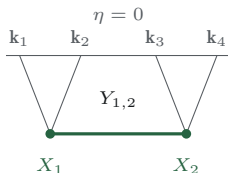
Symbols: total compatibility.

Functions: quadrangular polylogarithms.

The wavefunction and graph energies

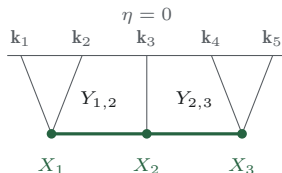
Conformally coupled ϕ^3 theory in dS [Arkani-Hamed et al. '17; Hillman '19]

$$\Psi[\varphi] = \exp \left[\sum_{N \geq 2} \frac{1}{N!} \int_{\mathbf{k}_1 \cdots \mathbf{k}_N} \psi_N \varphi_{\mathbf{k}_1} \cdots \varphi_{\mathbf{k}_N} \right], \quad X_v = \sum_{a \text{ at } v} |\mathbf{k}_a|, \quad Y_{i,j} = \left| \sum \mathbf{k}_a \right|.$$



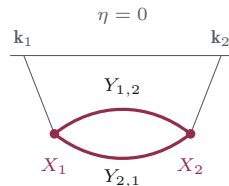
2-site chain

$$\begin{aligned} X_1 &= |\mathbf{k}_1| + |\mathbf{k}_2|, \\ X_2 &= |\mathbf{k}_3| + |\mathbf{k}_4|, \\ Y_{1,2} &= |\mathbf{k}_1 + \mathbf{k}_2|. \end{aligned}$$



3-site chain

$$\begin{aligned} X_1 &= |\mathbf{k}_1| + |\mathbf{k}_2|, \\ X_2 &= |\mathbf{k}_3|, \quad X_3 = |\mathbf{k}_4| + |\mathbf{k}_5|, \\ Y_{1,2} &= |\mathbf{k}_1 + \mathbf{k}_2|, \\ Y_{2,3} &= |\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3|. \end{aligned}$$



2-site bubble

$$\begin{aligned} X_1 &= |\mathbf{k}_1|, \\ X_2 &= |\mathbf{k}_2|, \\ Y_{1,2} &= |\ell|, \\ Y_{2,1} &= |\mathbf{k}_1 + \ell|. \end{aligned}$$

Polylogarithms in dS

$\psi_G(X_v, Y_{i,j})$: Tree-level wavefunction /Loop integrand in dS space
dS wavefunction from flat space

$$\psi_G(X_v, Y_{i,j}) = \int_0^\infty \prod_v dx_v \psi_G^{\text{flat}}(X_v + x_v, Y_{i,j})$$

Rational flat-space wavefunction $\psi_G^{\text{flat}} \rightarrow$ weight- n iterated logarithmic integrals.

Two-site chain [w. L. Ferro, T. Łukowski, et al.]

$$r = \frac{2Y_{1,2}}{X_1 + Y_{1,2}}, \quad s = \frac{2Y_{1,2}}{X_2 + Y_{1,2}}, \quad t = \frac{2Y_{1,2}(X_1 + X_2)}{(X_1 + Y_{1,2})(X_2 + Y_{1,2})}.$$

$$\psi_2 = \underbrace{[\text{Li}_2(r) - \text{Li}_1(r) \log r]}_{F_2(r)} + \underbrace{[\text{Li}_2(s) - \text{Li}_1(s) \log s]}_{F_2(s)} - \underbrace{[\text{Li}_2(t) - \text{Li}_1(t) \log t]}_{F_2(t)}.$$

ψ_2 is written as three copies of F_2 .

This expression has no explicit π term and vanishes in the soft limit $Y_{1,2} \rightarrow 0$.

Symbol alphabet and A -type cluster algebra

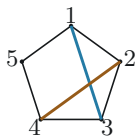
Two-site alphabet $\Leftrightarrow$ Plücker coordinates on $\text{Gr}(2, 5) = A_2$ [Capuano et al. '25; Mazloumi, Xu '25]

$$\psi_2 = F_2(r) + F_2(s) - F_2(t), \quad Y \equiv Y_{1,2}.$$

$$\begin{aligned} \mathcal{S}(\psi_2) &= \frac{2Y}{X_1 + Y} \otimes \frac{X_1 - Y}{X_1 + Y} \\ &+ \frac{2Y}{X_2 + Y} \otimes \frac{X_2 - Y}{X_2 + Y} \\ &- \frac{2Y(X_1 + X_2)}{(X_1 + Y)(X_2 + Y)} \otimes \frac{(X_1 - Y)(X_2 - Y)}{(X_1 + Y)(X_2 + Y)}. \end{aligned}$$

$$C_2 = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & X_1 + Y & X_1 - Y & X_1 + X_2 & 1 \end{pmatrix}.$$

$$\Delta_{ij} = \det(C_i, C_j); \quad \Delta_{i5} = 1 \text{ fixes the normalization.}$$



Coordinate	Letter
Δ_{12}	$X_1 + Y$
Δ_{23}	$-2Y$
Δ_{34}	$X_2 + Y$
Δ_{13}	$X_1 - Y$
Δ_{24}	$X_2 - Y$
Δ_{14}	$X_1 + X_2$

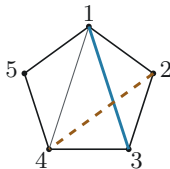
n -site chain $\Leftrightarrow A_{2n-2}$: triangulations of a $(2n + 1)$ -gon.

Total compatibility in type A

Compatibility means non-crossing chords.

SYM: Cluster adjacency

$$\Delta_{13} \otimes \Delta_{14} \otimes \Delta_{24}$$



Adjacent pairs are compatible; the first and third chords cross.

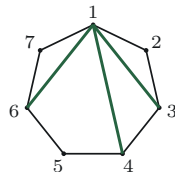
For chains: one type A cluster for the whole word.

Total compatibility also has a formulation for general graphs. [w. M. Capuano, L. Ferro et al.]

The cross-ratio uplift preserves total compatibility. [w. L. Ferro, T. Łukowski, et al.]

Cosmology: Total compatibility

$$\Delta_{13} \otimes \Delta_{14} \otimes \Delta_{16}$$



Every pair is compatible: all chords fit in one triangulation.

Plan of the talk

- Analytic structure & Total compatibility_[see Mattia and Allie's talk]
- Chain graphs & A -type QLi
- Loop graphs & B -type QLi

Functions with total compatibility

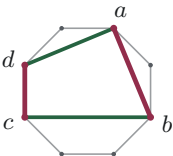
Quadrangular polylogarithms [Rudenko '20; Matveikin, Rudenko '22]

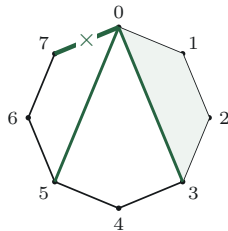
quadrangulations of **even** polygons $P \rightarrow \text{weight-}w \text{ QLi}_w^\pm(P) \rightarrow \text{totally compatible symbols}$

A sum over Quadrangulations

$$\text{QLi}_w = \sum_{\text{Quadrangulations}} \text{weight-}w \text{ MPL function}(\text{quadrangulation})$$

Quadrangular cross-ratio

$$q_{abcd} = \frac{\Delta_{ab} \Delta_{cd}}{\Delta_{bc} \Delta_{da}}$$


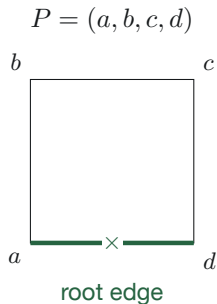


$$P = (0, 1, 2, 3, 4, 5, 6, 7)$$

This quadrangulation contributes

$$- \text{Li}_{1,1,1}(q_{0567}, q_{0345}, q_{0123}).$$

Start with a quadrilateral



One classical polylogarithm

$$\mathrm{QLi}_k^+(a, b, c, d) = (-1)^k \mathrm{Li}_k(q_{abcd}),$$

$$\mathrm{QLi}_k^-(a, b, c, d) = (-1)^{k+1} \mathrm{Li}_k(1/q_{abcd}).$$

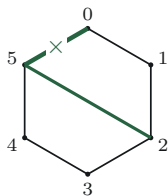
Even starting vertex: QLi^+ .

Odd starting vertex: QLi^- .

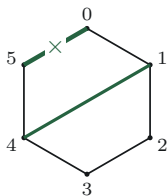
Examples: $\mathrm{QLi}_1^+(a, b, c, d) = \log(1 - q_{abcd}), \quad \mathrm{QLi}_1^-(a, b, c, d) = -\log(1 - 1/q_{abcd}),$
 $\mathrm{QLi}_2^+(a, b, c, d) = \mathrm{Li}_2(q_{abcd}).$

A hexagon: three quadrangulations

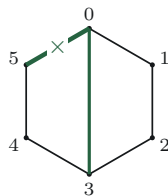
$$\mathrm{QLi}_2^+(0, 1, 2, 3, 4, 5) =$$



$$+ \mathrm{Li}_{1,1}(q_{0125}, q_{2345})$$



$$- \mathrm{Li}_{1,1}(q_{0145}, 1/q_{1234})$$



$$+ \mathrm{Li}_{1,1}(q_{0345}, q_{0123})$$

[Rudenko '20; Matveikin, Rudenko '22]

Decomposition of an n -site chain

QLi as compatible building blocks; their combination ψ_n ?

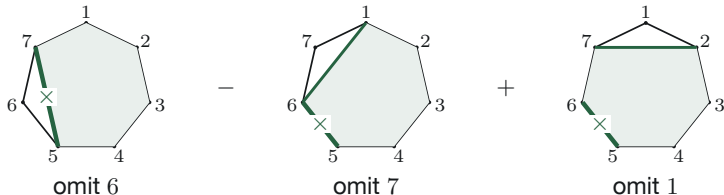
$$\psi_2 = F_2(5, 1, 2, 3) - F_2(4, 1, 2, 3) + F_2(4, 5, 2, 3).$$

$$F_2(0, 1, 2, 3) = \text{Li}_2(q_{0,1,2,3}) - \text{Li}_1(q_{0,1,2,3}) \log q_{0,1,2,3}$$

$$= \text{QLi}_2^+(0, 1, 2, 3) + \text{QLi}_1^+(0, 1, 2, 3) \log q_{0,1,2,3}.$$

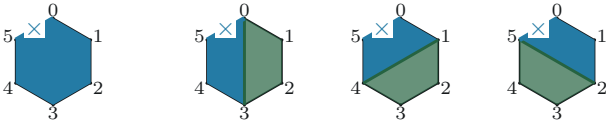
n -site chain, embedded in a $(2n + 1)$ -gon [w. L. Ferro, T. Łukowski, et al.]

$$\psi_n = F_n(2n + 1, 1, \dots, 2n - 1) - F_n(2n, 1, \dots, 2n - 1) + F_n(2n, 2n + 1, 2, \dots, 2n - 1).$$



$$F_n(0, 1, 2, \dots, 2n - 1) \equiv (-1)^n \text{QLi}_n^+(0, 1, 2, \dots, 2n - 1) \pmod{\text{products}}.$$

Product terms from Dissection

$$F_3(P) =$$


$$\begin{aligned}
 & -\Phi_3 = -[\text{QLi}_3^+(P) + \text{QLi}_2^+(P) \log q_P] & +\Phi_2(0345) & +\Phi_2(0145) & +\Phi_2(0125) \\
 & & \text{QLi}_1^+(0123) & \text{QLi}_1^-(1234) & \text{QLi}_1^+(2345)
 \end{aligned}$$

$q_P = q_{0123}q_{0345}$. [w. L. Ferro, T. Łukowski, et al.]

$$\Phi_r(S) = \text{QLi}_r^+(S) + \text{QLi}_{r-1}^+(S) \log q_S, \quad |S| = 2r,$$

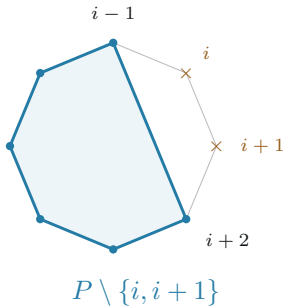
$$(-1)^n F_n(P) = \sum_{D=(S_0, \dots, S_k)} (-1)^k \Phi_{|S_0|/2}(S_0) \prod_{j=1}^k \text{QLi}_{|S_j|/2-1}^{(-1)^{S_j(1)}}(S_j).$$

Sum over even-face dissections D : S_0 contains the root $\times$; $S_1, \dots, S_k$ are the other faces.

Differential equation for F_n

From physics: Differential equation of F [w. L. Ferro, T. Łukowski, et al.]

$$dF_n(P) = \sum_{i=0}^{2n-1} F_{n-1}(P \setminus \{i, i+1\}) d\text{QLi}_1^{(-1)^{i-1}}(i-1, i, i+1, i+2).$$

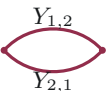


Boundary Value: Soft limit $F_n \rightarrow 0$ as an internal energy $Y_{i,i+1} \rightarrow 0$.

Plan of the talk

- Analytic structure & Total compatibility_[see Mattia and Allie's talk]
- Chain graphs & A -type QLi
- Loop graphs & B -type QLi

The symbol of the two-site bubble



$$X_1 \text{ --- } X_2 \quad X = X_1 + X_2$$

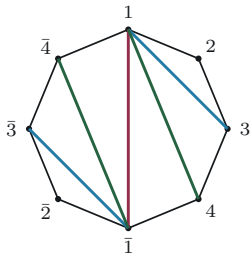
$$\begin{aligned} \mathcal{S}(\psi_2^\circ(X_1, Y_{1,2}, X_2, Y_{2,1})) &= \frac{X_1 + Y_{1,2} + Y_{2,1}}{X} \otimes \frac{(X_2 + Y_{1,2} + Y_{2,1})(X_2 - Y_{1,2} - Y_{2,1})}{(X_2 - Y_{1,2} + Y_{2,1})(X_2 + Y_{1,2} - Y_{2,1})} \\ &+ \frac{X + 2Y_{1,2}}{X} \otimes \frac{(X_1 + Y_{1,2} - Y_{2,1})(X_2 + Y_{1,2} - Y_{2,1})}{(X_1 + Y_{1,2} + Y_{2,1})(X_2 + Y_{1,2} + Y_{2,1})} \\ &+ \frac{X_2 + Y_{1,2} + Y_{2,1}}{X} \otimes \frac{(X_1 + Y_{1,2} + Y_{2,1})(X_1 - Y_{1,2} - Y_{2,1})}{(X_1 - Y_{1,2} + Y_{2,1})(X_1 + Y_{1,2} - Y_{2,1})} \\ &+ \frac{X + 2Y_{2,1}}{X} \otimes \frac{(X_1 - Y_{1,2} + Y_{2,1})(X_2 - Y_{1,2} + Y_{2,1})}{(X_1 + Y_{1,2} + Y_{2,1})(X_2 + Y_{1,2} + Y_{2,1})}. \end{aligned}$$

Symbol alphabet:

$$X_1 \pm Y_{1,2} \pm Y_{2,1}, \quad X_2 \pm Y_{1,2} \pm Y_{2,1}, \quad X + 2Y_{1,2}, \quad X + 2Y_{2,1}, \quad X.$$

The bubble diagram and cluster algebra B_3

Two sites $\longrightarrow$ a centrally symmetric octagon [Paranjape et al. '26]



$$(1, 3) \sim (\bar{1}, \bar{3}) \quad (1, 4) \sim (\bar{1}, \bar{4})$$

$$(1, \bar{1})$$

$$\Delta_{a,b} = \Delta_{\bar{a},\bar{b}}, \quad \Delta_{a\bar{b}} = 1 - \Delta_{ab}, \quad \Delta_{a\bar{a}} = 1.$$

$$\Delta_{1,3} = \frac{X_1 - Y_{1,2} + Y_{2,1}}{X}, \quad \Delta_{2,4} = \frac{X_2 - Y_{1,2} + Y_{2,1}}{X},$$

$$\Delta_{1,4} = \frac{X + 2Y_{2,1}}{X}, \quad \Delta_{1,\bar{2}} = \frac{X_2 - Y_{1,2} - Y_{2,1}}{X},$$

$$\Delta_{3,\bar{4}} = \frac{X_1 - Y_{1,2} - Y_{2,1}}{X}, \quad \Delta_{2,\bar{3}} = \frac{X + 2Y_{1,2}}{X},$$

$$\Delta_{1,\bar{3}} = \frac{X_2 + Y_{1,2} - Y_{2,1}}{X}, \quad \Delta_{2,\bar{4}} = \frac{X_1 + Y_{1,2} - Y_{2,1}}{X},$$

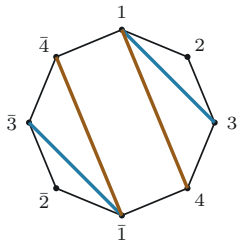
$$\Delta_{1,\bar{4}} = \frac{-2Y_{2,1}}{X}.$$

n -gon loop $\longleftrightarrow B_{2n-1}$: centrally symmetric $4n$ -gon.

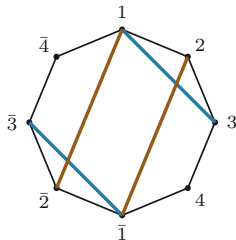
Compatibility in type B : the B_3 example

Include chord and its centrally symmetric partner.

Compatible: Δ_{13} and Δ_{14}



Incompatible: Δ_{13} and $\Delta_{1\bar{2}}$



No crossings among the two chord pairs;

Can appear together in a word

$(1, 3)$ crosses $(\bar{1}, \bar{2})$;

Do not appear together in a word

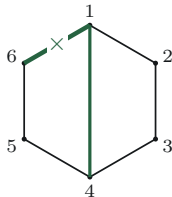
Total compatibility[w. M. Capuano, L. Ferro et al.]: one B -type cluster for each symbol word.

All chord pairs fit in one centrally symmetric triangulation.

Loops suggest a new B -type QLi

Chains

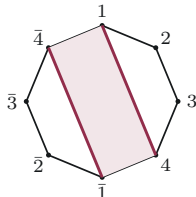
A -type alphabet $\longrightarrow$ ordinary polygons



Quadrangulations and A -type QLi. [Rudenko '20; Matveiakin, Rudenko '22]

n -gon

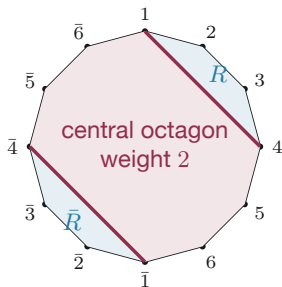
B_{2n-1} -type alphabet $\longrightarrow$ centrally symmetric $4n$ -gon



Centrally symmetric **partial quadrangulations** and B -type QLi. [w. L. Ferro, T. Łukowski, S. Paranjape et al., to appear.]

We define a new B -type QLi function, generalizing Rudenko's construction.

Weights in the B -type construction



$n = 3, m = 2$: weight $2 + 1 = 3$.

[w. L. Ferro, T. Łukowski, S. Paranjape et al., to appear.]

Partial quadrangulation of a $4n$ -gon

One self-conjugate central $4m$ -gon,
 $1 \leq m \leq n$.

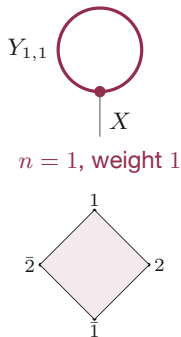
The complement is quadrangulated in $n - m$
conjugate pairs.

Central root: weight m .

Each ordinary pair: weight 1.

Total weight: $m + (n - m) = n$.

The tadpole (B_1) calls for a square root



First, the wavefunction is

$$\Delta_{12} = \frac{X + 2Y_{1,1}}{X}, \quad \psi_1^\circ = \log \frac{X + 2Y_{1,1}}{X} = \log \Delta_{12}.$$

The central quadrilateral supplies a square:

$$q_{\text{center}} = q_{12\bar{1}\bar{2}} = \left(\frac{\Delta_{12}}{\Delta_{1\bar{2}}} \right)^2.$$

We want to write

$$\psi_1^\circ = \text{Li}_1(1/w_{12}) \equiv -\text{QLi}_1^B(1, 2, \bar{1}, \bar{2}).$$

where

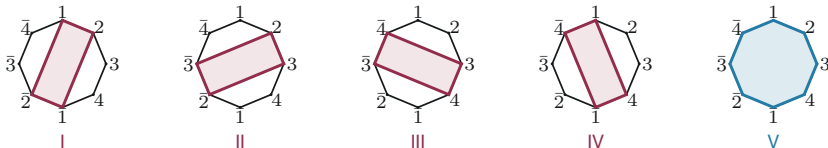
$$w_{ab} = -\frac{\Delta_{ab}}{\Delta_{a\bar{b}}}, \quad 1 - \frac{1}{w_{ab}} = \frac{1}{\Delta_{ab}}.$$

This minus sign keeps $1 - \frac{1}{w}$ in the cluster alphabet.

The opposite sign gives a new letter $\frac{2 - \Delta_{ab}}{\Delta_{a\bar{b}}}$.

The tadpole selects the square root; the alphabet fixes its sign.

The bubble: five geometric contributions



$$\begin{aligned} \text{QLi}_2^B(P) = & \underbrace{\text{Li}_{1,1}(q_{234\bar{1}}, 1/w_{12})}_{\text{I}} - \underbrace{\text{Li}_{1,1}(1/q_{34\bar{1}\bar{2}}, w_{23})}_{\text{II}} + \underbrace{\text{Li}_{1,1}(q_{4\bar{1}\bar{2}\bar{3}}, 1/w_{34})}_{\text{III}} - \underbrace{\text{Li}_{1,1}(1/q_{1234}, 1/w_{14})}_{\text{IV}} \\ & + \underbrace{\text{Li}_2(1/w_{1234})}_{\text{V}}. \end{aligned}$$

I–IV: weights $1 + 1$, $\text{Li}_{1,1}$.

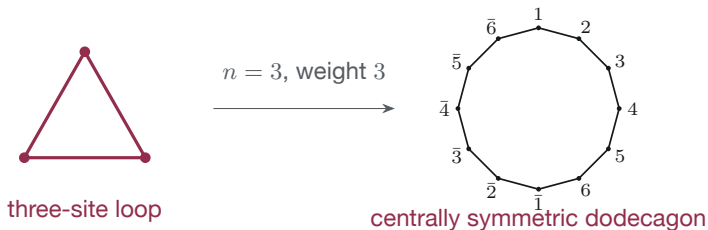
V: central octagon, weight 2, Li_2 .

$$\psi_2^\circ = \text{QLi}_2^B(1, 2, 3, 4, \bar{1}, \bar{2}, \bar{3}, \bar{4})$$

A single function for an n -gon

The n -gon wavefunction is a B -type QLi. [w. L. Ferro, T. Łukowski, S. Paranjape et al., to appear.]

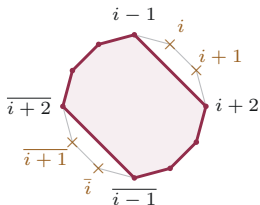
$$\psi_n^\circ = \text{QLi}_n^B(1, \dots, 2n, \bar{1}, \dots, \overline{2n}), \quad n \geq 2.$$



Differential equation for loops

Differential equation ($n \geq 2$) [w. S. He, X. Jiang, J. Liu, Q. Yang]

$$d\psi_n^\circ = \sum_{i=1}^{2n} \psi_{n-1}^\circ \left(\widehat{i}, \widehat{i+1} \right) d\text{QLi}_1^{(-1)^{i-1}}(i-1, i, i+1, i+2).$$



$$P' = P \setminus \{i, i+1, \bar{i}, \bar{i+1}\}.$$

$4n \rightarrow 4(n-1)$ vertices.

Central symmetry and cyclic order are preserved.

Boundary value: Soft limit $\psi_n^\circ \rightarrow 0$ as any $Y_{i,i+1} \rightarrow 0$.

This fixes the additive constant to be 0.

The B -type construction satisfies the differential equation and boundary data.

[w. L. Ferro, T. Łukowski, S. Paranjape et al., to appear.]

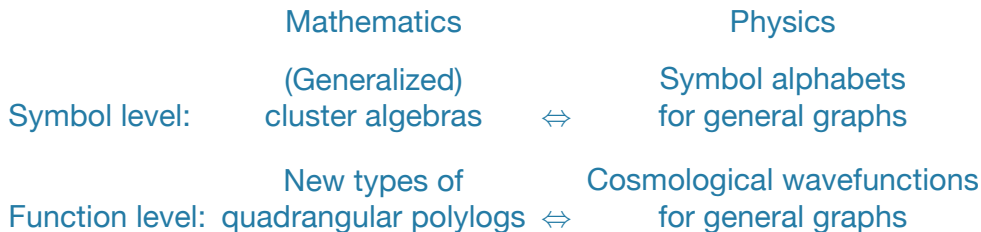
Chains and loops

	Chains [w. L. Ferro, T. Łukowski, et al.]	<i>n</i> -gon loops [w. L. Ferro, T. Łukowski, S. Paranjape et al., to appear.]
Alphabet	Type <i>A</i>	Type <i>B</i>
Geometry	Rooted even polygons	Centrally symmetric $4n$ -gons
Root	A chosen boundary edge	The central face
Functions	QLi + product completion	New <i>B</i> -type QLi

Total compatibility refines the function space;
differential equations and boundary data fix the wavefunction.

Outlook

- Towards general graphs:



- Correlators: conformal ϕ^3 , massive and spin
- General FRW / higher weights

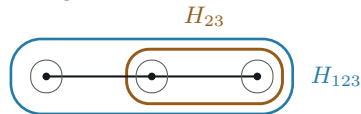
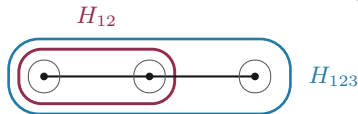
Backup: the flat-space tubing formula

$$\psi_G^{\text{flat}}(X, Y) = N_G \sum_{\tau \in \mathcal{B}_G^{\text{max}}} \prod_{T \in \tau} \frac{1}{H_T}, \quad N_G = \prod_{e \in E_G} 2Y_e.$$

A **maximal binary tubing** τ is a set of nested or vertex-disjoint connected subgraphs to which no further compatible tube can be added.

$$H_T = \sum_{v \in V_T} X_v + \sum_{e \in E_G} n_{e,T} Y_e, \quad n_{e,T} = \text{number of cut ends of } e.$$

Three-site chain: two maximal tubings. The root and all singleton tubes are included.



$$\frac{\psi_{P_3}^{\text{flat}}}{N_{P_3}} = \frac{1}{H_1 H_2 H_3 H_{12} H_{123}}$$

$$+ \frac{1}{H_1 H_2 H_3 H_{23} H_{123}}$$

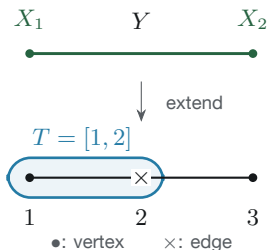
$$H_{12} = X_1 + X_2 + Y_{2,3}, \quad H_{23} = X_2 + X_3 + Y_{1,2}, \quad H_{123} = X_1 + X_2 + X_3.$$

Glew '25, Sec. 2: N_G is the normalization used here; stripping it leaves the tubing sum.

Backup: tubes, H_T , and Δ

Two-site chain: tubes are connected intervals of the extended graph.

$$(t_1, t_2, t_3) = (X_1 + Y, -2Y, X_2 + Y), \quad H_T = \sum_{a \in T} t_a, \quad Y = Y_{1,2}.$$



Tube T	H_T	Coordinate
$\{1\}$	$X_1 + Y$	Δ_{12}
$\{2\}$	$-2Y$	Δ_{23}
$\{3\}$	$X_2 + Y$	Δ_{34}
$\{1, 2\}$	$X_1 - Y$	Δ_{13}
$\{2, 3\}$	$X_2 - Y$	Δ_{24}
$\{1, 2, 3\}$	$X_1 + X_2$	Δ_{14}

$$C_a = (1, z_a)^T, \quad z_a = \sum_{k=1}^{a-1} t_k \quad (a = 1, \dots, 4), \quad \Delta_{ab} = \det(C_a, C_b).$$

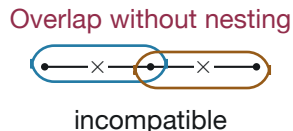
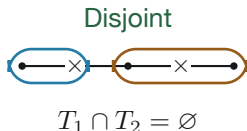
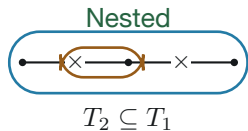
$$H_{[i,j]} = t_i + \dots + t_j = z_{j+1} - z_i = \Delta_{i,j+1}.$$

$1 \leq i \leq j \leq 3$; $C_5 = (0, 1)^T$ fixes $\Delta_{a5} = 1$ for $a \leq 4$. [w. L. Ferro, T. Łukowski, et al.]

Backup: the ordered single-cluster condition

Two constraints on each symbol word [w. M. Capuano, L. Ferro et al.]

Compatibility Every pair of tubes is nested or disjoint.



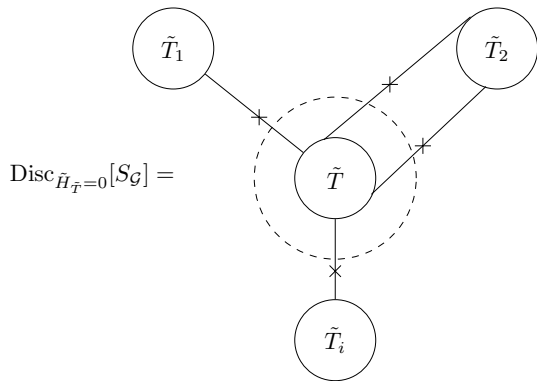
Ordering A larger tube appears before a tube contained in it.

$$T_1 \supset T_2 \implies \cdots \otimes H_{T_1} \otimes \cdots \otimes H_{T_2} \otimes \cdots$$

The discontinuity structure gives these constraints for any graph.
They are the conditions used in the full bootstrap table.

Backup: compatibility from discontinuities

A discontinuity resolves the graph into connected components.



[w. M. Capuano, L. Ferro et al., Fig. 4]

For a connected tube T :

$$H_T = \sum_{v \in T} X_v + \sum_{(i,j) \text{ leaving } T} Y_{i,j}.$$

The cut separates T from the components $T_1, T_2, \dots$

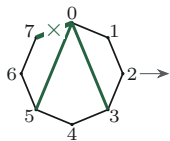
Further discontinuities must respect these components.

Compatible tubes

$\longleftrightarrow$ compatible letters.

Tube compatibility is defined on the extended graph.

Backup: from polygons to MPLs



Quadrangulation $\longrightarrow$ rooted dual tree $\longrightarrow$ weighted words.

$$\text{Li}_0^R([\varphi_1, m_1] \otimes \cdots \otimes [\varphi_d, m_d]) = \text{Li}_{m_1, \dots, m_d}(\varphi_1, \dots, \varphi_d).$$

At a node, $t = t_1 \star t_2 \star t_3$ is the quasi-shuffle of descendant words.

$$t_Q = \begin{cases} -[q, 1] \otimes t, & \text{even root,} \\ [1/q, 1] \otimes t + [1/q, 1] \cdot t, & \text{odd root,} \end{cases} \quad T_P = \sum_{Q \in \mathcal{Q}(P)} t_Q.$$

[Rudenko '20] [w. L. Ferro, T. Łukowski, et al.]

Contraction: $[\varphi, m] \cdot [\psi, r] = [\varphi\psi, m + r]$.

$$\text{Li}_1^R(W) = - \sum_{j=1}^d m_j \text{Li}_{m_1, \dots, m_j+1, \dots, m_d}(\varphi_1, \dots, \varphi_d).$$

A $2n$ -gon has $n - 1$ quadrilaterals: $\text{Li}_0^R(T_P)$ gives QLi_{n-1} ; $\text{Li}_1^R(T_P)$ gives QLi_n .

Empty subtrees: $1 \star t = t$, $[\varphi, m] \cdot 1 = 0$.

Backup: the full B -type definition

Sum over symmetric partial quadrangulations: central $4m$ -gon S_0 , ordinary quadrilaterals elsewhere.
Rotate S_0 so $S_0(1)$ is odd.

$$\mathrm{cr}^B(S_0) = (-1)^{m+1} \frac{\prod_{j=1}^m \Delta_{S_0(2j), S_0(2j+1)}}{\prod_{j=1}^m \Delta_{S_0(2j+1), S_0(2j+2)}}.$$

[w. L. Ferro, T. Łukowski, S. Paranjape et al., to appear.]

For each ordinary node, with descendant product t :

$$t_Q^{\mathrm{ord}} = \begin{cases} t \otimes [q, 1] + t \cdot [q, 1], & \text{even root,} \\ -t \otimes [1/q, 1], & \text{odd root.} \end{cases}$$

$$\mathrm{QLi}_n^B(P) = \mathrm{Li}_0^R \left[\sum_{Q \in \mathcal{Q}^\sigma(P)} (-1)^n \left(\star_{j \geq 1} t_{S_j}^{\mathrm{ord}} \right) \otimes [\mathrm{cr}^B(S_0), m] \right].$$

The central weight m plus $n - m$ ordinary weights gives n .

Backup: opening the loop at tube level

Two-site example: $H_I = \sum_{a \in I} T_a$, $T_4 = \Lambda \rightarrow \infty$; T_1, T_2, T_3 fixed.



Avoid 4: finite chain tubes. Contain 4: $H_I = \Lambda + \text{finite}$.

$H_{34}/H_4 \rightarrow 1$; the common $\log \Lambda$ terms cancel in the full symbol.

$$\lim_{\Lambda \rightarrow \infty} \mathcal{S}(\psi_2^\circ) = \frac{H_1}{H_{123}} \otimes \frac{H_3}{H_{23}} + \frac{H_3}{H_{123}} \otimes \frac{H_1}{H_{12}} = -\mathcal{S}(\psi_2).$$

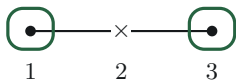
•: vertex gap; x: edge gap. For an n -gon, the cut removes label $2n$.

Backup: tubes and inclusion–exclusion

Subtract a whole word when all its tubes miss an endpoint.

Two-site example: endpoints 1 and 3; one word from each group in S_M .

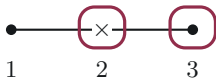
Keep: both endpoints



$$+H_1 \otimes H_3$$

Different tubes, both ends.

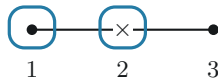
Subtract: misses 1



$$-H_2 \otimes H_3$$

Group S_R .

Subtract: misses 3



$$-H_2 \otimes H_1$$

Group S_L .

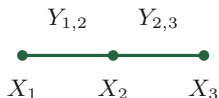
Add back words missing both ends: this group is zero.

Here the last-slot tubes $\{1\}$, $\{1, 2\}$, $\{2, 3\}$, $\{3\}$ all touch an endpoint.

$$\underbrace{S_M - S_R - S_L + 0}_{\text{both endpoints covered}} = -\mathcal{S}(\psi_{\text{chain}}) \implies \boxed{\mathcal{S}(\psi_{\text{chain}}) = S_L - S_M + S_R}.$$

S_L, S_M, S_R are the three F -symbols. The drawings show tube supports of ordered words.

Backup: three-site symbol



Weight 3

13 letters

104 terms [Hillman '19]

The alphabet, organized by connected subgraphs [w. L. Ferro, T. Łukowski, et al.]

$$\mathcal{A}_3 = \left\{ \begin{aligned} &X_1 + X_2 + X_3, \\ &X_1 \pm Y_{1,2}, \quad X_3 \pm Y_{2,3}, \\ &X_2 \pm Y_{1,2} \pm Y_{2,3}, \\ &X_1 + X_2 \pm Y_{2,3}, \\ &X_2 + X_3 \pm Y_{1,2} \end{aligned} \right\}.$$

Both signs chosen independently.

Can we organize these letters before integrating?